

JavaScript Assignment — Loops (Real-World Practice)

Topic: Loops in JavaScript

(Each problem carries equal marks. Write clean code with comments & output screenshots.)

Section 1 — For Loop (5 Problems)

Q1. Print your institute's name 5 times using a for loop.

Sample Input:

```
name = "CJC Karve Nagar"
```

Sample Output:

```
CJC Karve Nagar  
CJC Karve Nagar  
CJC Karve Nagar  
CJC Karve Nagar  
CJC Karve Nagar
```

Q2. Display all even roll numbers between 1 to 20.

Sample Input: —

(No input required)

Sample Output:

```
2 4 6 8 10 12 14 16 18 20
```

Q3. Calculate the total price of items in a shopping cart.

Sample Input:

```
cart = [120, 80, 250, 50]
```

Sample Output:

```
Total Amount = 500
```

Q4. Print the marks and average of 5 subjects.

Sample Input:

```
marks = [65, 72, 81, 54, 90]
```

Sample Output:

```
Total Marks = 362  
Average = 72.4
```

Q5. Generate and display the multiplication table of an employee ID.

Sample Input:

```
empId = 7
```

Sample Output:

```
7 x 1 = 7  
7 x 2 = 14  
...  
7 x 10 = 70
```

Section 2 — While Loop (5 Problems)

Q1. Print numbers 1 to 10 using a while loop.

Sample Input: —

(No input required)

Sample Output:

```
1 2 3 4 5 6 7 8 9 10
```

Q2. Count how many employees have salary above ₹50,000.

Sample Input:

```
salaries = [48000, 52000, 60000, 43000, 75000]
```

Sample Output:

```
Employees with salary > 50000 = 3
```

Q3. Display reverse of an invoice number.

Sample Input:

invoice = 4567

Sample Output:

Reverse = 7654

Q4. Calculate total digits in a customer's mobile number.

Sample Input:

mobile = 9876543210

Sample Output:

Total Digits = 10

Q5. Check if a customer's coupon code is a palindrome.

Sample Input:

code = 1221

Sample Output:

Valid Coupon! (Palindrome)

Section 3 — Do...While Loop (5 Problems)

Q1. Display a countdown for an online test timer (from 5 to 1).

Sample Input: —

(No input required)

Sample Output:

```
5
4
3
2
1
Start Test!
```

Q2. Keep accepting item prices until user enters 0, then display total.

Sample Input:

```
100, 50, 70, 0
```

Sample Output:

```
Total Bill = 220
```

Q3. Print all odd order numbers from 1 to n.

Sample Input:

```
n = 15
```

Sample Output:

```
1 3 5 7 9 11 13 15
```

Q4. Display power consumption calculation ($\text{unit}^{\text{rate}}$) until limit is reached.

Sample Input:

```
unit = 2, rate = 5
```

Sample Output:

```
Power = 32
```

Q5. Keep asking for a password until it matches the correct one.

Sample Input:

```
User attempts: 1234 → 5678 → 9999 (Correct)
```

Sample Output:

```
Access Denied  
Access Denied  
Access Granted
```

Section 4 — For...Of Loop (5 Problems)

Q1. Display each student name from a list.

Sample Input:

```
students = ["Amit", "Kunal", "Sneha", "Raj"]
```

Sample Output:

```
Amit  
Kunal  
Sneha  
Raj
```

Q2. Calculate the total salary of employees.

Sample Input:

```
salaries = [25000, 30000, 40000]
```

Sample Output:

```
Total Salary = 95000
```

Q3. Display all vowels from a company name.

Sample Input:

```
name = "TechMahindra"
```

Sample Output:

```
Vowels: e, a, i, a
```

Q4. Find the highest marks from an array using for...of loop.

Sample Input:

```
marks = [45, 67, 89, 56]
```

Sample Output:

```
Highest Marks = 89
```

Q5. Count how many employees have name length greater than 4.

Sample Input:

```
employees = ["Raj", "Kunal", "Pooja", "Om", "Neha"]
```

Sample Output:

```
Names with length > 4 = 2
```

Section 5 — For...In Loop (5 Problems)

Q1. Display all keys of a student record object.

Sample Input:

```
const student = { name: "Sneha", age: 22, course: "React" };
```

Sample Output:

```
name  
age  
course
```

Q2. Display all key-value pairs of a product.

Sample Input:

```
const product = { id: 101, name: "Mouse", price: 450 };
```

Sample Output:

```
id : 101  
name : Mouse  
price : 450
```

Q3. Count total properties in a customer object.

Sample Input:

```
const customer = { id: 11, name: "Rahul", city: "Pune", mobile: 9876 };
```

Sample Output:

```
Total Properties = 4
```

Q4. Copy all properties from one object to another using for...in loop.

Sample Input:

```
const obj1 = { course: "Java", duration: "6 months" };
```

Sample Output:

```
New Object: { course: "Java", duration: "6 months" }
```

Q5. Print only numeric property values from a mixed object.

Sample Input:

```
const data = { a: 10, b: "Hello", c: 40, d: "JS" };
```

Sample Output:

```
10  
40
```
